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STANDARD OPERATING PROCEDURE

TITLE:	OPERATIONS, LITHIUM ION BATTERY INCIDENT RESPONSE
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REPLACES:	NONE
ACCOUNTABILITY:	OPERATIONS
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Purpose:

To establish procedures to safely interact with hybrid and electric vehicles, Battery Energy Storage Systems (BESS), and other products that use Lithium Ion battery systems at incidents.

General Information:

- I. All Operations personnel must be competent in performing the procedures established in this publication.
- II. The Fire Department is responsible for managing Lithium Ion Battery related incidents in the City of Fort Worth and, by way of approved mutual/automatic aid agreements, in neighboring cities as well.

It should be noted that incidents involving Lithium Ion Vehicles and Battery Energy Storage Systems are highly hazardous, toxic and volatile situations. Lithium-ion battery fires release toxins such as carbon monoxide, hydrogen cyanide, hydrogen fluoride and cobalt. These toxins are both an inhalation and absorption hazard. PPE and SCBA shall be utilized on any event involving lithium ion batteries or systems. In addition to standard fires, which require fuel, heat, and oxygen to continue burning, lithium-ion (Li-ion) battery cells can experience a chemical reaction known as thermal runaway, which does not require oxygen or a visible flame, if it occurs within a tightly packed cell assembly. This process can continue to drive up temperature and pressure until the battery cell ruptures, which in turn can release explosive gases. These incidents should be handled as HAZMAT incidents. All appropriate HAZMAT documents and forms should be filled out when these incidents occur.

Instructions/Procedures

- 1 Battalion Chief, 3 apparatus (one engine, one truck/quint, and one HAZMAT company) will be dispatched on all open-air Lithium Ion fires to include, but not limited to, roadway, parking lot, open field, etc.
- One alarm assignment and a HAZMAT company will be dispatched on all structure fires, to include, but not limited to, residential, commercial, parking garage, underground structures and fires below railway trestles or roadway bridges.

III. This SOP should be applied, by all responding crews, when interacting with hybrid and electric vehicles, Battery Energy Storage Systems (BESS), and other products that use Lithium Ion battery systems for any of the following conditions:

- Stranded or Disabled Vehicle
- Motor Vehicle Accident
- Vehicle Smoke or Fire (Refer to the attached ESA Thermal Event FF Safety Checklist)
- Flooded Vehicle (Refer to the attached ESA Flood Safety Towing Attachment)
- Medical Emergency involving vehicle occupants still in the vehicle
- Structure Fire with vehicle exposure
- Structure Fire with vehicle involvement
- Proximity Fire with vehicle Exposure
- Charging Malfunction with vehicle plugged in
- Battery Energy Storage System (BESS) smoke, fire or other issues (Refer to the attached ESA Thermal Event FF Safety Checklist)
- Electric Bikes, Scooters or other products that use lithium ion battery systems that are producing smoke or fire
- Towing a lithium ion powered vehicle that has been in an accident or involved in a flooding or fire event. (Refer to the attached ESA First Responder RAP)

IV. List of attachments

- QR Code ESA App
- ESA Thermal Event FF Safety Checklist
- ESA Flood Safety Towing Checklist
- ESA First Responder RAP
- ESA Towing Checklist

V. List of Equipment and Resources Required or Recommended

The following is a list of required items for safely interacting with hybrid and electric vehicles, Battery Energy Storage Systems (BESS), and other products containing lithium ion batteries:

- CO Meter as a minimum, 4 Gas Meter preferred.
- Full Structural PPE with SCBA (when signs of thermal runaway, gases or other products of combustion, electrical damage, submersion, vehicle rollover or heavy physical damage are present). This is required.
- Thermal Imaging Camera
- Assess Battery Compartment/Battery Pack Damage

VI. The following is a list of situation and environment dependent equipment recommendations:

- Portable Ground Monitor

- Engine Apparatus
- Attack Lines
- Supply Lines
- Manifold or Gated Wye
- Hand Tools – 6' Z Hook
- Wheel Chocks
- Stabilization Wedges and Cribbing
- EV Charging Port Plugs
- EV Fire Blanket(s)
- Aerial Apparatus
- Traffic Management Devices or Apparatus
- High Strength Extrication Tools
- Vented Overpack Drums or Containers
- Pigs, Dikes, Absorbent Sheets
- Granular Absorbent
- Ventilation Fans
- Cable or Chain Hoists
- Ratchet Straps
- Vehicle Stabilization Struts

VII. Documentation

- All appropriate ICS Forms and FWFD HAZMAT forms should be filled out.
 1. ICS 201 Incident Briefing
 2. ICS 202 Incident Objectives
 3. ICS 203 Organization Assignment List
 4. ICS 204 Assignment List
 5. ICS 205 Incident Radio Communications
 6. ICS 206 Medical Plan
 7. ICS 208HM Site Safety and Control Plan
 8. ICS 214 Activity Log
 9. FWFD 307 Hazardous Materials Personnel Exposure Report
 10. FWFD 715 Hazardous Materials Incident-Transportation
 11. FWFD 719 Chemical Exposure Record (Roster)
 12. FWFD 720 Fire Department Incident Review Form
 13. FWFD 726 Hazardous Materials Incident-Facility
 14. HAZMAT Pre-Entry Form
 15. Pre-Entry Brief
 16. Entry Log
 17. Air Management Log
 18. DECON Log
 19. Debrief Form

VIII. Definitions

- **EV/BEV:** Electric vehicle/battery electric vehicle-Any vehicle that uses electric motors, either in full or in part, as propulsion.
- **HEV:** Hybrid electric vehicle-A car that integrates a small battery and an electric motor to enhance the efficiency of the engine. The ICE (internal combustion engine) maintains the battery's charge. It cannot be charged by plugging into an electrical supply. Hybrids can travel short distances on electric power only.
- **PHEV:** Plug-in hybrid electric vehicle-A vehicle configured like a traditional hybrid, but with a bigger battery pack that can be charged by plugging into an electrical source. This vehicle can start a trip on electric-only propulsion, after being plugged in, and can usually travel further on battery-only than a HEV.
- **Li-ion/LITHIUM-ION BATTERY:** The current chemistry standard in electric vehicle batteries which offers the needed energy density, power, and fast charging capability.
- **1st RESPONDER CUT LOOP/CUT LOOP/FIREFIGHTER CUT LOOP:** A low voltage harness loop that, when cut, is designed to isolate the high voltage power to just the battery pack.
- **BATTERY CELL:** The smallest unit of the battery pack. They are individual single cells designed as either a cylindrical, pouch, or prismatic or combination depending on the manufacturer. If the cylindrical battery pack housing is damaged these cells can become dislodged or even ejected in a severe collision.
- **BATTERY MODULE:** A module can contain several hundred battery cells. The module design allows for electronic battery management thermal control and allows for greater capacity by using space more efficiently.
- **BATTERY PACK:** All the components of energy storage for a vehicle. The pack contains all the modules (which include the individual cells) used to store energy. Depending on the vehicle manufacturer and vehicle type the battery pack can be used as a structural member of the vehicle's body on many cars. Additionally, on most EVs the battery pack is located on the bottom of the vehicle and will run the length of the car between the axles. On HEVs and PHEVs, the battery pack location varies but mostly could be found towards the vehicle's rear (under or behind the back seat or in the trunk).
- **PV:** Photovoltaic system
- **DC/DC Converter:** Converts Voltage from one range to another.
- **DRIVETRAIN:** System of components used to transfer power from the engine/motor to the wheels.
- **REGENERATIVE BRAKING:** Energy recovery mechanism that slows down a moving vehicle by converting its kinetic energy into a form that can be either used immediately or stored until needed. This occurs during deceleration (coasting) and/or active braking.
- **BATTERY MANAGEMENT SYSTEM(BMS):** A system of components used to monitor and control the charge, discharge, and temperature of the battery cells.
- **Thermal Runaway:** Thermal runaway is a chain reaction within a battery cell which an increase in the temperature of a lithium-ion battery can cause a further increase in temperature, causing a exothermic reaction leading to the release of flammable gas and the potential for an explosion. It is caused by the pressure created from the build-up of heat in the battery, which can be due to mechanical, electrical, or thermal abuse. Things like excessive current, internal short circuits, external heat sources, puncturing, or flame impingement.
- **CO (Carbon Monoxide):** Is a colorless, poisonous, odorless, tasteless, flammable gas that is slightly less dense than air. CO is an incomplete product of combustion.

- **H (Hydrogen):** Is colorless, odorless, tasteless, non-toxic, and highly combustible gas. H is a natural element found in water and air
- **RAP (Risk Assessment Placard):** This is an assessment sticker, from the ESA, which provides the level of risk, provided by the ESA, for proper handling and management of the vehicle.
- **AUTHORITY HAVING JURISDICTION (AHJ):** an organization, office, or individual responsible for enforcing the requirements of a code or standard, or for approving equipment, materials, an installation, or a procedure.
- **BESS:** Battery Energy Storage System
- **ESS:** Energy Storage System

Instructions/Procedures:

I. VEHICLE IDENTIFICATION

- **INITIAL CALL:** FAO shall make every effort to identify the vehicle as a hybrid or electric vehicle while processing the initial call and inform responding companies. If vehicle make, model, and manufacturing year can be ascertained it **shall also be** communicated to responding companies. Call Type shall be adjusted to reflect increased threat level of hybrid and electric vehicles.
- **UPON ARRIVAL:** Establish safe work zone through apparatus positioning and proper traffic management tools.
- Identify or confirm Hybrid Electric Vehicle through formal and/or informal assessment. Perform initial size up and adjust safe work zone if needed based on vehicle presentation. Any indications of heavy physical damage, smoke/gas/fire, submersion, or exposure to heat or fire should require safe work zone of 75' minimum and full PPE with SCBA.
- Perform 360 Degree survey at a minimum distance of 20' from the vehicle with a gas meter and thermal imaging camera to detect the presence of CO/LEL or the visual indication of off gassing. Heavily damaged vehicles may produce displaced batteries or other high voltage equipment that should be identified and avoided initially.
- **CONTACT THE ESA: 855 ESA SAFE (1-855-372-7233) and initiate on scene support.**
- Prioritize life safety needs followed by property conservation and establish an action plan.

Note: If occupants are still in the vehicle and gas/smoke/fire are present, perform rapid extrication.

II. Response involving fire with no exposure

- 1 Truck/Quint, 1 Engine, 1 Satellite Hazmat company, 1 Battalion Chief

Response involving fire with exposure

- 1 Truck/Quint, 1 Engine, 1 Satellite Hazmat company, 1 Battalion Chief

Response involving a fire within a residential structure

- Standard 1 alarm residential assignment, 1 Hazmat company

Response involving a fire within a commercial structure

- Standard commercial alarm assignment, 1 Hazmat company

III. IMPLEMENT ACTION PLAN (HAZARD MANAGEMENT)

- Position appropriate wheel chocks to prevent vehicle movement. Hybrid and Electric Vehicles may present without audible indications of being energized or in drive mode and have the potential to accelerate at rapid rates. Crews should use extreme caution when approaching the vehicle until the vehicle has been fully secured.
- Attempt to gain access to the passenger cockpit through door handle access. If electromechanical door handles do not present and do not operate the door, use appropriate glass removal techniques to gain access through either of the front windows. Follow ESA guidance, **855 ESA SAFE (1-855-372-7233)**, to actuate the interior manual door release overrides and open the door(s).
- If the doors cannot be opened, visually assess the cockpit dash display for illumination and signs of vehicle status. If the vehicle appears to be on, attempt to locate a power button or ignition key and turn the vehicle off resulting in INITIAL SHUTDOWN. If key fobs, vehicle cards, or cell phones are present, remove those items and isolate in faraday bags, if available, or a minimum of 30' from the involved vehicle. If the passenger cockpit cannot be accessed and the vehicle cannot be turned off due to obstruction or other scenario, progress to the following step.
- Follow ESA, **855 ESA SAFE (1-855-372-7233)**, guidance to locate, access and perform the vehicle energy isolation procedure as needed. Minor to moderately damaged vehicles may not require any actions other than vehicle shut down. SECONDARY SHUTDOWN Procedures may be required and will be specific to the vehicle involved. If specific vehicle guidance is not being utilized, access the low voltage battery and the vehicle First Responder Cut loop if present and disconnect or cut the negative battery cable or plug and cut or disconnect the Fireman's Loop.
- Some vehicle, and special circumstances may require removal of fuses, relays, or manual service disconnects. High Voltage Service Disconnects should only be performed with

proper guidance from the ESA and appropriate PPE. Arc Flash potential is always present when interacting with high voltage components.

- An EV Plug should be used, if available, to confirm the vehicle is in Neutral/Park and the Parking Brake is engaged. **Note: An EV Plug does not de-energize the vehicle, it only communicates with the vehicle and confirms the vehicle is in Neutral/Park and the Parking Brake is engaged via a visual light indicator on the EV Plug**

Note: DO NOT CUT, INADVERTENTLY COMPRESS OR CONTACT ORANGE HIGH VOLTAGE CABLES OR VEHICLE COMPONENTS WITH ELECTRICAL HAZARD WARNING LABELS. DO NOT CUT FIRST RESPONDER LOOPS OR BATTERY CABLES WHEN VEHICLE PRESENTATION DOES NOT MERIT ENERGY ISOLATION

IV. IMPLEMENT ACTION PLAN (VEHICLE STABILIZATION)

- Apply four points of contact under the A and C pillars of the vehicle if the vehicle is on all four wheels. Avoid deep contact points that would apply point load pressure to the floor pan of the vehicle. This could result in physical abuse of the battery pack.
- Rollover or collision resulting in unnatural resting position of the vehicle will require advanced stabilization techniques. Avoid contact with the floor pan of the vehicle that could result in point load pressure and physical abuse of the battery pack. Back tie techniques and less reliable contact points such as control arms, wheels, and suspensions may be required as anchor points to avoid battery pack abuse.
- Focal points for undercarriage contact should be the OEM's engineered lifting points located near the frame rails and rockers under the A pillars and Posterior (B or C) pillars.
- An EV Plug should be used, if available, to confirm the vehicle is in Neutral/Park and the Parking Brake is engaged. **Note: An EV Plug does not de-energize the vehicle, it only communicates with the vehicle and confirms the vehicle is in Neutral/Park and the Parking Brake is engaged via a visual light indicator on the EV Plug**

V. IMPLEMENT ACTION PLAN (FIRE OPERATIONS)

- If the vehicle is off gassing, smoking or actively burning, identify the source of the event. Battery-based fires will produce products of combustion through the engineered vent points of the battery pack or the paths of least resistance. This will typically originate from the vehicle undercarriage.
- Incomplete combustion of Li-ion battery gas contains toxic gasses that may result in serious injury or death if inhaled. Full PPE and SCBA shall always be worn in these conditions.
- Water shall be the primary suppression application. Multi class extinguishing agents designed to encapsulate or emulsify shall only be deployed in the absence of water. Traditional foam should be avoided.

- DO NOT APPLY WATER to early off-gassing battery packs that are not combusting.
- ALLOW THE VEHICLE TO BURN if the following conditions exist:
 1. Fully involved fire will not create an imminent life safety hazard.
 2. Exposures can be adequately protected.
- Consider Suppression efforts if the above conditions are not met.
- Ensure adequate water supply is available to support fire suppression operations. Continuous flow may be required for up to eight hours or longer.
- Rapid attack lines should be deployed initially for rapid extrication attempts or aggressive offensive attacks designed to knock the initial fire down.
- Extended suppression efforts will require appropriate water supply, primary and secondary lines, back up lines, and remote water monitors.
- Effective suppression efforts require Direct Cooling of the Li-ion batteries. This is only accomplished with water penetration to the interior of the battery pack enclosure. Controlled streams should be directed into the vent points or breaches in the battery pack.
- Reposition the vehicle for vehicle lifting and stabilization to optimize access to the vent or breach points of the battery pack.
- Repositioning of the vehicle through lifting and stabilization techniques will require streams to be directed at the fire for the purpose of protecting personnel and equipment in the direct path of the fire and gas.
- Suppression operations may produce much higher volumes of non-combusted gas and environmental/atmospheric management should be implemented.
- Suppression operations may also produce water shed that is hazardous and requires containment and mitigation. Appropriate environmental management should be implemented.
- Adequate suppression and extinguishment will be confirmed by a continuous decline in thermal condition of the battery pack as well as the decline of CO in the off-gassing atmosphere. This is not an indication of absolute containment. Stranded energy may still be present in the vehicle with unvented battery cells and secondary thermal runaway is highly probable.
- Vehicles should not be transferred down the chain of custody or relocated until stable conditions are present for a minimum of one hour.
- Movement of the vehicle, post incident, also creates a high probability of secondary thermal event. Movement of the vehicle post incident by responding crews or other responders (Tow and Recovery professionals) should be guided by the ESA in accordance with SAE J2990 Risk Assessment recommendations. (Refer to the attached ESA Towing Checklist)

VI. IMPLEMENT ACTION PLAN (SUBMERGED OR FLOOD DAMAGED VEHICLES)

- Vehicles fully submerged should only be accessed by qualified public safety or salvage divers.
- ESA guidance, **855 ESA SAFE (1-855-372-7233)**, should be sought for dive-based access, rigging, and removal of the submerged vehicle.
- Electric Vehicles may experience thermal events while submerged. This will be identified through the presence of gas emanating from the surface of the water and active fire bursts under the surface of the water. Divers and First Responders should not enter the water if the vehicle is experiencing an active event while submerged.

- Once the vehicle is removed from the water, it should be propped up to allow water to drain from the vehicle. The vehicle should remain in a static position and stable conditions should be present for a minimum of one hour prior to secondary movement or transfer through the chain of custody. There is a high probability that the vehicle will experience a thermal event upon removal from the water. First responders should establish fire operations action plans upon vehicle removal. (Refer to the attached ESA Flooded Safety Towing Checklist)

VII. POST INCIDENT MANAGEMENT

- Responding companies shall properly label the vehicle with the ESA RAP sticker (Refer to the attached ESA First Responder RAP) and require tow and recovery professionals to complete compliant ESA Risk Assessments. (Refer to the attached ESA Towing Checklist)
- Battery components or debris shall only be managed by First Responder personnel. Thermal Imaging Cameras, four gas meters, and Full PPE and SCBA shall be utilized when handling separated high voltage materials. Non-conductive tools such as plastic scoop shovels should be used to secure separated materials.
- Damaged batteries may be unvented and fully energized and pose significant risk potential to First and Second (towing personnel) Responders. If displaced batteries appear unvented, thermal imagery should be used to detect cell temperature. Any isolated cells exceeding 200 degrees F should be cooled immediately with water through vessel submersion or direct flow. Seek ESA guidance if the scenario presents itself.
- ESA guidance, **855 ESA SAFE (1-855-372-7233)**, shall be adhered to when performing material clean-up operations.
- This process includes DOT and EPA compliant handling, removal and transport of battery materials and should only be performed with ESA guidance or Haz Mat Specialist equivalent.

VIII. SPECIAL CONSIDERATIONS VEHICLE CONFINEMENT (PARKING GARAGES, PARKING LOTS, VEHICLE ENCLOSURES)

- These environments may require rapid isolation of combustible gasses and fire to prevent fire spread to exposures and accumulation of combustible and toxic gas. Gas clouds (smoke) emanating from EV's should always be perceived as highly toxic and highly combustible until determined otherwise through atmospheric monitoring.
- First Responders should consider deployment of EV Fire blankets on the involved vehicle in accordance with the manufacturer's recommendations. Fire Blankets should be deployed as temporary containment tools that are not designed to stop thermal runaway. This may need to be done out and away from the fire to strategically get in front of the fire. First Responders should be prepared to manage fuel rich gas products that have developed under the blanket when the blanket is removed. This removal process may result in combustion of the released gasses and should be addressed with water stream application.
- Adjacent exposures such as other vehicles or mobile objects can then be relocated away from the fire source.
- Structural elements such as pre and post-tensioned concrete decks, walls, or ceilings made of combustible materials should be aggressively protected with fire streams to reduce the risk of collapse and fire spread. It is theoretically possible for structural elements such as, but not limited to, columns and beams to fail, possibly resulting in total collapse of the structure.

- Ventilation and gas monitoring may be necessary to safely manage atmospheric conditions.
- Consider hydraulic ventilation tactics through hand line applications to produce venturi effects and add water weight to the gas in highly pressurized environments with unfavorable natural ventilation.
- **COMMERCIAL VEHICLES (BUSES, SEMI TRACTORS, COMMERCIAL VANS, BOX TRUCKS)**
 - Contact the ESA **855 ESA SAFE (1-855-372-7233)**, for guidance regarding vehicle interaction requirements. Commercial vehicle systems have more complexities related to integrated suppression and detection systems, braking, and energy isolation. Additionally, battery packs may be located on the roofs of the vehicle and highly reinforced compartments. Roof top packs may require elevated streams for proper suppression applications.
 - USE CAUTION when accessing or opening compartments that contain Li-ion batteries. Combustible and toxic gas may be contained in a fuel rich state within these compartments and produce highly combustible/explosive conditions when accessed. All compartment spaces that may potentially be compromised should be monitored with a gas meter probe in FULL PPE and SCBA prior to opening.
 - Commercial Vehicle electric parking brakes may be compromised due to damage and First and Second Responders should apply appropriate wheel chocks and potential anchoring to prevent undesired vehicle movement.
 - Full PPE and SCBA shall be worn.

IX. CHARGING STATIONS (LEVEL I, II Residential and LEVEL II, III Commercial)

- Electrical vehicles that are plugged in to a charging station and require First Responder interaction should be unplugged or disconnected from the charging station. Call the ESA for guidance. **855 ESA SAFE (1-855-372-7233)**
- In the case of fire, off gas, smoke, audible signs of electrical damage, submersion or collision while plugged in, charging stations should be isolated remotely through charging station electric panels, E Stops, or identified island breakers. Lock out/tag out procedures should be followed.

X. STRUCTURE FIRE (Hybrid and Electric Vehicles as an exposure and fire source)

- Tactical priorities should include the mitigation of fire spread to the garage or area where the electric vehicle is located. This may include defensive fire stream positioning or vehicle removal/relocation.
- Responding companies should exercise extreme caution when making entry on residential structure fires where the hybrid or electric vehicle is involved. The presentation of dense, white smoke (gas) should be an immediate indicator to consider the presence of li-ion battery gas. This gas has high concentrations of CO and H which may result in flash over or gas explosion. This can produce intense fire, shrapnel and flying debris up to and including complete overhead door displacement.

XI. DAMAGED VEHICLE COLLECTION POINTS (Salvage Yards, Tow Yards, Evidence Impound Lots)

All damaged hybrid and electric vehicles should be isolated with 50' boundaries from all combustibles or with isolation barriers on three sides. These barrier specifications are set in accordance with SAE J2990 to prevent large scale conflagrations or exposure and life safety hazards within the jurisdiction.

XII. Battery Energy Storage Systems (Fixed Facilities), Semi-Mobile Systems, and Battery Storage Rooms of Fixed Structures

Battery energy storage systems (BESS) pose special hazards to firefighters. With recent advances in battery technology and renewable energy, li-ion batteries have become one of the leading solutions for large-scale energy storage. Buildings or facilities containing a BESS may not have markings that specifically identify the presence of these systems. Markings may only indicate a general electrical hazard is present. BESS may be found in basements, or in floor in a multi-story building. An independent facility containing a BESS may appear to be a steel building resembling a shipping container or other smaller enclosures. These facilities may be found anywhere. They may be stand-alone or incorporated in another structure, including residential areas. The use of BESS to support the electrical grid is becoming standard and these systems will be found throughout the city.

The types of catastrophic failures that can occur in all battery systems are amplified by the size and scale of the BESS. Larger systems contain more energy and have the potential to create large volumes of toxic and flammable gases that can become explosive when contained. The hazards are dependent on the design of the BESS, characteristics of the compartments containing the BESS, and levels of fire protection systems in the structure.

In smaller residential settings, a li-ion battery module may undergo thermal runaway as a result of exposure to a heat source unrelated to the battery. For example, a battery module located in or near a garage that is exposed to a car fire can be damaged and has the potential to have a thermal runaway event. This may cause it to liberate toxic, flammable gases and lead to a potentially high energy ignition. Fires involving Li-Ion batteries have a very high heat release rate and present extinguishment challenges.

Stranded energy is residual energy within a Li-ion battery or BESS. This presents a significant fire, electrical shock, and/or explosion hazard to firefighters. The severity of the hazard is in direct relationship to the state of charge in the battery. Assume they are charged.

BESS failures can occur for a variety of reasons including but not limited to:

- Thermal abuse (external temperatures)
- Physical/Mechanical damage
- Electrical abuse (over-charging or repeated excessive charging rates)
- Environmental events (electrical surge, etc.)
- Internal faults within the battery cells
- Other electrical failures

Response to Battery Energy Storage Systems

- BESS must always be considered energized. Firefighters should exercise extreme caution when dealing with BESS and all energized electrical equipment.
- Request utility company to respond
- Do not make entry or approach BESS building or compartment. Introducing fresh air may result in a deflagration.
- Isolate the area. Recommended initial evacuation distance is 150 feet. Do not enter the fenced area. The exception to this is a savable life/known rescue.
- Be aware of explosion potential and off-gassing of hazardous materials. White colored smoke is a good indication of hazardous off-gassing.
- Upwind, Upstream and Uphill shall be utilized when possible.
- Place apparatus in a safe location away from BESS and overhead power lines.
- Protect exposures.
- DEFENSIVE FIREFIGHTING, water streams are the preferred agent for response to Li-ion battery fires (Li-ion is not water reactive).
- If a fire has not developed and only smoke is visible, take a defensive stance toward the system and be prepared to apply water spray to exposures.
- If a fire develops, take a defensive stance toward the burning BESS unit and apply water to neighboring battery enclosures and exposures.
- Maintaining a safe distance from the unit involved (large commercial systems, at least 150').
- Response crews should allow the battery to burn out. Water should be applied to adjacent battery enclosures and exposures.
- The Incident Commander will make the ultimate determination regarding hazard mitigation. The hazard mitigation plan should be developed in partnership with the utility representative and/or responsible party.
 1. Through this careful approach, hazardous materials technicians may take calculated steps to mitigate the hazard.
 2. Depending on the BESS type and size, mitigation steps may include identification of the hazard, separating it from electrical supply (i.e., electrical grid or photovoltaic system), ventilation, and cooling.
- Firefighters shall wear full personal protective equipment, including SCBA with face-piece.
- If identified shut off the unit/system by operating any visible disconnects or E-stops (shutting off the disconnect or E-stops does not remove the energy from the battery). To isolate any PV system and ESS in an emergency, multiple disconnects may need to be shut off.
- Li-ion batteries that are in thermal runaway or off gassing will create hazardous flammable and toxic atmospheres.
- Firefighters must stay out of the vapor cloud/smoke.
- Due to construction of the unit, thermal imaging cameras or infrared temperature sensors may not give true thermal conditions. But can be used to identify trending temperatures.
- Hazardous failure of batteries can occur from physical damage, exterior fire, or an internal malfunction of a cell. Smoke, popping or hissing sounds from an ESS system can be an indication of a hazardous condition. When batteries or cells enter thermal runaway, there is typically a period of smoke (may be under pressure). The smoke is flammable and may ignite at any time.

XIII. Responding to a Venting ESS Product

- Evacuate the area. Never open any doors or remove panels to ESS units.
- Maintaining a safe distance from the unit involved (large commercial systems, at least 150').
- Contact vendor-specific technical support for assistance including battery management system (BMS) data.
- For residential units that are located inside a dwelling unit or garage, the space should be properly ventilated with charged handlines in place (open garage door only if safe to do so).
- Maintain a safe distance from the ESS and monitor visually for signs of thermal runaway. A remote FDC may be present on larger commercial or utility ESS to support a sprinkler system inside the enclosure.
- Each manufacturer will have a recommended time for a battery pack to cool down. This can be near a full work cycle of 12 hours or more.
- Defensive firefighting, water spray is the preferred agent for response to Li-ion battery fires (Li-ion is not water reactive).
- If a fire has not developed and only smoke is visible, take a defensive stance toward the system and evaluate the need, and practical ability to safely apply water on exposures.
- If a fire develops, take a defensive stance toward the burning unit. Evaluate the need and practical ability to safely apply water on exposures.
- Response crews should allow the battery to burn out. Water should be applied to adjacent battery enclosures and exposures.
- Full PPE and SCBA shall be worn.

XIV. Types of Batteries

- **Lead Acid Batteries**

Lead Acid batteries are also used as BESS. These systems pose chemical burn hazard, inhalation hazard and the risk of contaminated runoff. If found upgrade the assignment to a hazardous materials assignment and operate electrical disconnects. Utilize full PPE, including SCBA, and direct water application from 40 feet (upwind if possible) can be applied to control a fire.

- **Lithium Ion Batteries**

Li-ion Batteries can produce heat, toxic and flammable gases, and contain stranded energy.

XV. RESIDENTIAL BESS

BESS are typically found as part of a modern photovoltaic system (PV) but may be present even in the absence of a photovoltaic system (PV). BESS systems can be Li-ion battery or lead acid battery based. Each of these poses a potential for toxic exposure, fire and thermal burns, and chemical exposure.

Emergency Procedures for Residential BESS systems.

If a residential BESS system is identified the following steps shall be taken:

- Notify Command
- Upgrade assignment to a Hazardous Materials Response, isolate the area.
- It is critical to operate electrical disconnects if it is safe to do so. This will isolate the battery system. After operation of the electrical disconnects it is important to recognize that the batteries will still pose an electrical hazard due to the stranded energy that they contain.
- Crews will wear full PPE and with SCBA Face Piece in place.
- A charged hose line will be in place.
- Isolate the area surrounding the residential BESS, minimum of 40 feet. Hazardous Materials crews will set a hot zone.
- Avoid parking apparatus directly in front of the house
- Evaluate if the Battery Pack has been impacted by the initial fire or other event
- Contact vendor-specific technical support for assistance including BMS data.
- Consider a monitoring/cooling period of 12 hours in which the battery is monitored for signs of thermal runaway. This is a passive cooling period without water being applied. Direct application of water is ineffective for cooling to prevent a thermal runaway due to the construction of the battery pack. Temperature readings should be taken as early as possible and repeated at regular intervals (10 minutes) to identify trends. Is the battery getting hotter or beginning to cool?
- Contact the responsible party. They will be informed that they must contact the manufacturer for technical support.
- Command Officers shall make attempts to notify the building owner or occupant that there is a risk associated of a secondary event including fire as a result of continued power generation from the PV panels.

- Command or their designee may make notification over the radio that this notification has been made.
- Water application is the recommended method of firefighting.
- Protect exposures through defensive firefighting.
- The Responsible Party must arrange for proper disposal and cleanup.
- Full PPE and SCBA shall be worn.

Note: All BESS contain quantities of hazardous materials. In the event of an emergency with a BESS, a toxic environment may be created that is not visible. Metering of the environment by hazardous materials crews is necessary as is the usage of full personal protective equipment.

XV. Hazardous considerations

- Signs of Thermal Runaway:
- Sight: White smoke or gas, Fire, Arcing
- Sound: Arcing, Hissing, or Popping of battery vent due to pressurized release of gas within the battery cells
- Smell: Electrical burning odor with a “sweet” undertone.
- Responding companies who are exposed to Li-Ion battery gas should be evaluated for potential toxic inhalation exposure and should be documented as a HAZMAT Exposure. (Refer to the Documentation Section in this SOP)
- PPE ensembles do not provide adequate protection against prolonged direct flame exposure to battery-based fire production. Battery Flames may extend 6-10 feet from the battery pack and will have a blowtorch presentation. Direct exposure may result in significant burns.
- Environmental conditions may be significantly impacted by incomplete combustion of Li-ion battery gas and careful risk/benefit analysis should be conducted before initiating suppression efforts. Indirect cooling or inadequate water supply and improper stream direction will likely extend the event significantly and produce a much larger impact environmentally and atmospherically.
- Direct contact with high voltage components may result in arcing events and electrocution.
- Secondary thermal events are highly probable, and First Responders should strongly consider escort operations for tow and recovery professionals transporting hybrid and electric vehicles that have been submerged or experienced a thermal event.
- Li-ion batteries may produce off gassing that cannot initially be seen with the naked eye. Thermal imaging cameras are essential to early visual detection of off gassing and gas meters indicating CO production and LEL.
- Li-ion batteries may ignite with little warning through rapid, pressurized releases of combustible gas. Responding companies should always wear FULL PPE with SCBA and

exercise extreme caution when interacting with heavily compromised hybrid and electric vehicles with hazardous presentations.

A handwritten signature in black ink, appearing to read 'J. Davis', written in a cursive style.

James Davis
FIRE CHIEF